

# Ratio Reasoning: Convert Measurements within the Same System

Lesson 3-6

**Name:** Type your name

**Date:** Today's date

**Class:** Class period

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## TODAY'S OBJECTIVES

**Content Objective:** I can use ratio reasoning to convert between units within the same measurement system.

**Language Objective:** I can explain a conversion using the words unit ratio, convert, measurement system, and per.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



## WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Convert Measurements within the Same System

Conversion factor

Unit ratio

Convert

Measurement system

Double number line

Ratio table



Tap any word to see what it means and a picture.

1

Feet into inches stays inside the  measurement system.

2

The ratio that trades one unit for another is a

.

3

A ratio comparing an amount to exactly  unit is a unit ratio.

4

To change a measurement to another unit without changing the amount is to

.

5

Inches and feet belong to the customary

 ; centimeters and meters to the metric one.

6

Two lines whose matching ticks show the same amount in two units form a

.

7

A table whose every column holds the same comparison is a

.

## Write About the Math

**How many cup-scoops of broth does the recipe need, and how do you know?**

*¿Cuántas tazas de caldo necesita la receta y cómo lo sabes?*

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Convert Measurements within the Same System**  
*Convertir medidas dentro del mismo sistema*

☐ **Conversion factor**  
*Factor de conversión*

☐ **Unit ratio**  
*Razón unitaria*

☐ **Convert**  
*Convertir*

☐ **Measurement system**  
*Sistema de medidas*

**Model:** Find the scale factor ( $120 \text{ divided by } 8 = 15$ ), then multiply each ingredient by it. — Students find the scale factor of 15 and explain that multiplying both ingredients by 15 keeps the recipe proportional (5 cups tomatoes becomes 75, 3 cups mozzarella becomes 45).

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- The unit ratio is \_\_\_\_ cups to 1 quart. Cups are \_\_\_\_ than quarts, so I \_\_\_\_ to convert.  $3 \times 4 =$  \_\_\_\_ cups. The amount of broth did not change — only the \_\_\_\_ did.
  - My answer is \_\_\_\_ because \_\_\_\_.
- Mi respuesta es \_\_\_\_ porque \_\_\_\_.*
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### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is \_\_\_\_.
- Mi afirmación es \_\_\_\_.*

- **My evidence is** \_\_\_\_, **so** \_\_\_\_.
- Mi evidencia es* \_\_\_\_, *entonces* \_\_\_\_.

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## Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

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☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.